Hotel Booking Cancellation Prediction

Introduction / Problem Overview

Hotel booking cancellations create important challenges for hotel management because they affect occupan planning, staffing, revenue stability, and overbooking decisions. When many reservations are canceled, hot lose revenue and face difficulty using rooms efficiently.

This project applies supervised machine learning classification methods to predict whether a booking will l canceled. The goal is to compare different models, evaluate their performance, and identify the algorithm t supports practical business decision-making.

Dataset Description

This project uses the Hotel Booking Demand dataset, which contains hotel reservation records for City Ho Resort Hotel bookings. Each row represents one booking, and the target variable, `is_canceled`, indic whether the reservation was canceled (`1`) or not (`0`).

The dataset includes a mixture of customer, reservation, and service-related information such as:

- booking lead time
- meal type
- market segment
- distribution channel
- average daily rate (ADR)
- number of special requests
- customer type
- room assignment details

This dataset is suitable for a binary classification task because the main objective is to predict a yes/no out from multiple features.

Project Objective

The objective of this project is to build and compare several supervised machine learning classification mo predict whether a hotel booking will be canceled.

More specifically, this analysis aims to:

- preprocess and clean the booking data

- explore variables related to cancellation behavior
- train and evaluate multiple classification models
- compare results using appropriate evaluation metrics
- select the model that best satisfies the project goal

The final outcome is a model and interpretation that can help hotels better understand cancellation patterr support decisions that reduce revenue loss.

Data Cleaning and Preprocessing

Before modeling, the dataset was cleaned and prepared carefully:

- duplicate rows were removed
- columns with strong leakage or too many missing values were dropped
- missing values in selected columns were filled using suitable methods
- categorical variables were encoded
- numerical features were scaled when needed for specific models

These preprocessing steps improved the quality of the dataset and made the model comparison more reliat

Code Documentation Notes

The code cells below include inline comments using `#` so each step of the workflow is easy to follow. These comments explain the purpose of the code for data cleaning, exploratory data analysis, preprocessing modeling, evaluation, and final comparison.

In [126…

```
# Import the main libraries for data analysis, visualization, and machine
import pandas as pd
import numpy as np
import sklearn.ensemble
import matplotlib.pyplot as plt
import seaborn as sns
```

In [127…

```
# Load the hotel booking dataset and preview the first rows.
data = pd.read_csv('hotel_bookings.csv')
data.head()
```

Out[127…

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_week_number	arr
0	Resort Hotel	0	342	2015	July	27	
1	Resort Hotel	0	737	2015	July	27	
2	Resort Hotel	0	7	2015	July	27	
3	Resort Hotel	0	13	2015	July	27	
4	Resort Hotel	0	14	2015	July	27	

5 rows × 32 columns

In [128…

```
# Check the number of rows and columns in the original dataset.
print('Dataset size is :', data.shape)
```

```
Dataset size is : (119390, 32)
```

In [129…

```
# Inspect column names, data types, and missing values.
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 32 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   hotel                           119390 non-null  object
 1   is_canceled                     119390 non-null  int64
 2   lead_time                       119390 non-null  int64
 3   arrival_date_year               119390 non-null  int64
 4   arrival_date_month              119390 non-null  object
 5   arrival_date_week_number        119390 non-null  int64
 6   arrival_date_day_of_month       119390 non-null  int64
 7   stays_in_weekend_nights         119390 non-null  int64
 8   stays_in_week_nights            119390 non-null  int64
 9   adults                          119390 non-null  int64
 10  children                        119386 non-null  float64
 11  babies                          119390 non-null  int64
 12  meal                            119390 non-null  object
 13  country                         118902 non-null  object
 14  market_segment                  119390 non-null  object
 15  distribution_channel            119390 non-null  object
 16  is_repeated_guest               119390 non-null  int64
 17  previous_cancellations          119390 non-null  int64
 18  previous_bookings_not_canceled  119390 non-null  int64
 19  reserved_room_type              119390 non-null  object
 20  assigned_room_type              119390 non-null  object
 21  booking_changes                 119390 non-null  int64
 22  deposit_type                    119390 non-null  object
 23  agent                           103050 non-null  float64
 24  company                         6797 non-null    float64
 25  days_in_waiting_list            119390 non-null  int64
 26  customer_type                   119390 non-null  object
 27  adr                             119390 non-null  float64
 28  required_car_parking_spaces     119390 non-null  int64
 29  total_of_special_requests       119390 non-null  int64
 30  reservation_status              119390 non-null  object
 31  reservation_status_date         119390 non-null  object
dtypes: float64(4), int64(16), object(12)
memory usage: 29.1+ MB
```

In [130…

```python
# Explore the distribution of countries in the dataset.
data['country'].value_counts()
```

Out[130…

```
country
PRT    48590
GBR    12129
FRA    10415
ESP     8568
DEU     7287
       ...
DJI        1
BWA        1
HND        1
VGB        1
NAM        1
Name: count, Length: 177, dtype: int64
```

In [131…

```python
# Check how many bookings are linked to each travel agent.
data['agent'].value_counts()
```

Out[131…

```
agent
9.0      31961
240.0    13922
1.0       7191
14.0      3640
7.0       3539
         ...
289.0        1
432.0        1
265.0        1
93.0         1
304.0        1
Name: count, Length: 333, dtype: int64
```

In [132…

```
# Count the number of unique values in each column.
data.nunique()
```

Out[132…

```
hotel                                 2
is_canceled                           2
lead_time                           479
arrival_date_year                     3
arrival_date_month                   12
arrival_date_week_number             53
arrival_date_day_of_month            31
stays_in_weekend_nights              17
stays_in_week_nights                 35
adults                               14
children                              5
babies                                5
meal                                  5
country                             177
market_segment                        8
distribution_channel                  5
is_repeated_guest                     2
previous_cancellations               15
previous_bookings_not_canceled       73
reserved_room_type                   10
assigned_room_type                   12
booking_changes                      21
deposit_type                          3
agent                               333
company                             352
days_in_waiting_list                128
customer_type                         4
adr                                8879
required_car_parking_spaces           5
total_of_special_requests             6
reservation_status                    3
reservation_status_date             926
dtype: int64
```

In [133…

```
# View the frequency of each hotel type.
data['hotel'].value_counts()
```

Out[133…

```
hotel
City Hotel      79330
Resort Hotel    40060
Name: count, dtype: int64
```

In [134…

```
# Check whether the 'babies' column contains unusual values.
data['babies'].value_counts()
```

Out[134…

```
babies
0     118473
1        900
2         15
10         1
9          1
Name: count, dtype: int64
```

In [135…

```
# Inspect the 'company' column before deciding whether to keep or drop it
data['company'].value_counts()
```

Out[135…

```
company
40.0     927
223.0    784
67.0     267
45.0     250
153.0    215
        ...
104.0      1
531.0      1
160.0      1
413.0      1
386.0      1
Name: count, Length: 352, dtype: int64
```

In [136…

```
# View the number of bookings in each market segment.
data['market_segment'].value_counts()
```

Out[136…

```
market_segment
Online TA        56477
Offline TA/TO    24219
Groups           19811
Direct           12606
Corporate         5295
Complementary      743
Aviation           237
Undefined            2
Name: count, dtype: int64
```

In [137…

```
# View the number of bookings in each distribution channel.
data['distribution_channel'].value_counts()
```

Out[137…

```
distribution_channel
TA/TO        97870
Direct       14645
Corporate     6677
GDS            193
Undefined        5
Name: count, dtype: int64
```

In [138…

```
# Count duplicate rows in the dataset.
data.duplicated().sum()
```

Out[138…

```
np.int64(31994)
```

In [139…

```python
# Remove duplicate rows to avoid repeated observations in the analysis.
data.drop_duplicates(inplace=True)
```

In [140…

```python
# Confirm the new dataset shape after removing duplicates.
data.shape
```

Out[140… (87396, 32)

In [141…

```python
# Recheck the dataset structure after removing duplicates.
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 87396 entries, 0 to 119389
Data columns (total 32 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   hotel                           87396 non-null  object
 1   is_canceled                     87396 non-null  int64
 2   lead_time                       87396 non-null  int64
 3   arrival_date_year               87396 non-null  int64
 4   arrival_date_month              87396 non-null  object
 5   arrival_date_week_number        87396 non-null  int64
 6   arrival_date_day_of_month       87396 non-null  int64
 7   stays_in_weekend_nights         87396 non-null  int64
 8   stays_in_week_nights            87396 non-null  int64
 9   adults                          87396 non-null  int64
 10  children                        87392 non-null  float64
 11  babies                          87396 non-null  int64
 12  meal                            87396 non-null  object
 13  country                         86944 non-null  object
 14  market_segment                  87396 non-null  object
 15  distribution_channel            87396 non-null  object
 16  is_repeated_guest               87396 non-null  int64
 17  previous_cancellations          87396 non-null  int64
 18  previous_bookings_not_canceled  87396 non-null  int64
 19  reserved_room_type              87396 non-null  object
 20  assigned_room_type              87396 non-null  object
 21  booking_changes                 87396 non-null  int64
 22  deposit_type                    87396 non-null  object
 23  agent                           75203 non-null  float64
 24  company                         5259 non-null   float64
 25  days_in_waiting_list            87396 non-null  int64
 26  customer_type                   87396 non-null  object
 27  adr                             87396 non-null  float64
 28  required_car_parking_spaces     87396 non-null  int64
 29  total_of_special_requests       87396 non-null  int64
 30  reservation_status              87396 non-null  object
 31  reservation_status_date         87396 non-null  object
dtypes: float64(4), int64(16), object(12)
memory usage: 22.0+ MB
```

In [142…

```python
# Drop columns with too many missing values or leakage from the future ta
#Dropping company and agent just for missing values and 'reservation_stat
data.drop(columns=['company', 'agent', 'reservation_status', 'reservation
```

In [143…

```python
# Confirm the new dataset shape after dropping columns.
data.shape
```

Out[143… (87396, 28)

In [144…
```
# Fill missing values in 'children' with the most frequent value (mode).
data['children'].fillna(data['children'].mode()[0], inplace=True)
data['children'].shape
```

```
C:\Users\persh\AppData\Local\Temp\ipykernel_16684\104712648.py:1: FutureWarni
value is trying to be set on a copy of a DataFrame or Series through chained
assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work b
the intermediate object on which we are setting values always behaves as a co

For example, when doing 'df[col].method(value, inplace=True)', try using
'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) in
to perform the operation inplace on the original object.


  data['children'].fillna(data['children'].mode()[0], inplace=True)
```

Out[144… (87396,)

In [145…
```
# Fill missing values in 'country' with the most frequent country.
data['country'] = data['country'].fillna(data['country'].mode()[0])
data['country'].shape
```

Out[145… (87396,)

In [146…
```
# Verify that missing values have been handled correctly.
data.isna().sum()
```

Out[146…
```
hotel                             0
is_canceled                       0
lead_time                         0
arrival_date_year                 0
arrival_date_month                0
arrival_date_week_number          0
arrival_date_day_of_month         0
stays_in_weekend_nights           0
stays_in_week_nights              0
adults                            0
children                          0
babies                            0
meal                              0
country                           0
market_segment                    0
distribution_channel              0
is_repeated_guest                 0
previous_cancellations            0
previous_bookings_not_canceled    0
reserved_room_type                0
assigned_room_type                0
booking_changes                   0
deposit_type                      0
days_in_waiting_list              0
customer_type                     0
adr                               0
required_car_parking_spaces       0
total_of_special_requests         0
dtype: int64
```

Exploratory Data Analysis

Before training the models, exploratory data analysis was performed to understand how booking features r cancellation behavior. The following charts and pairplots highlight patterns that may be useful for classific such as lead time, deposit type, market segment, customer behavior, and service requests.

In [147…

```
# Check the class balance of the target variable.
data['is_canceled'].value_counts(normalize=True)
```

Out[147…

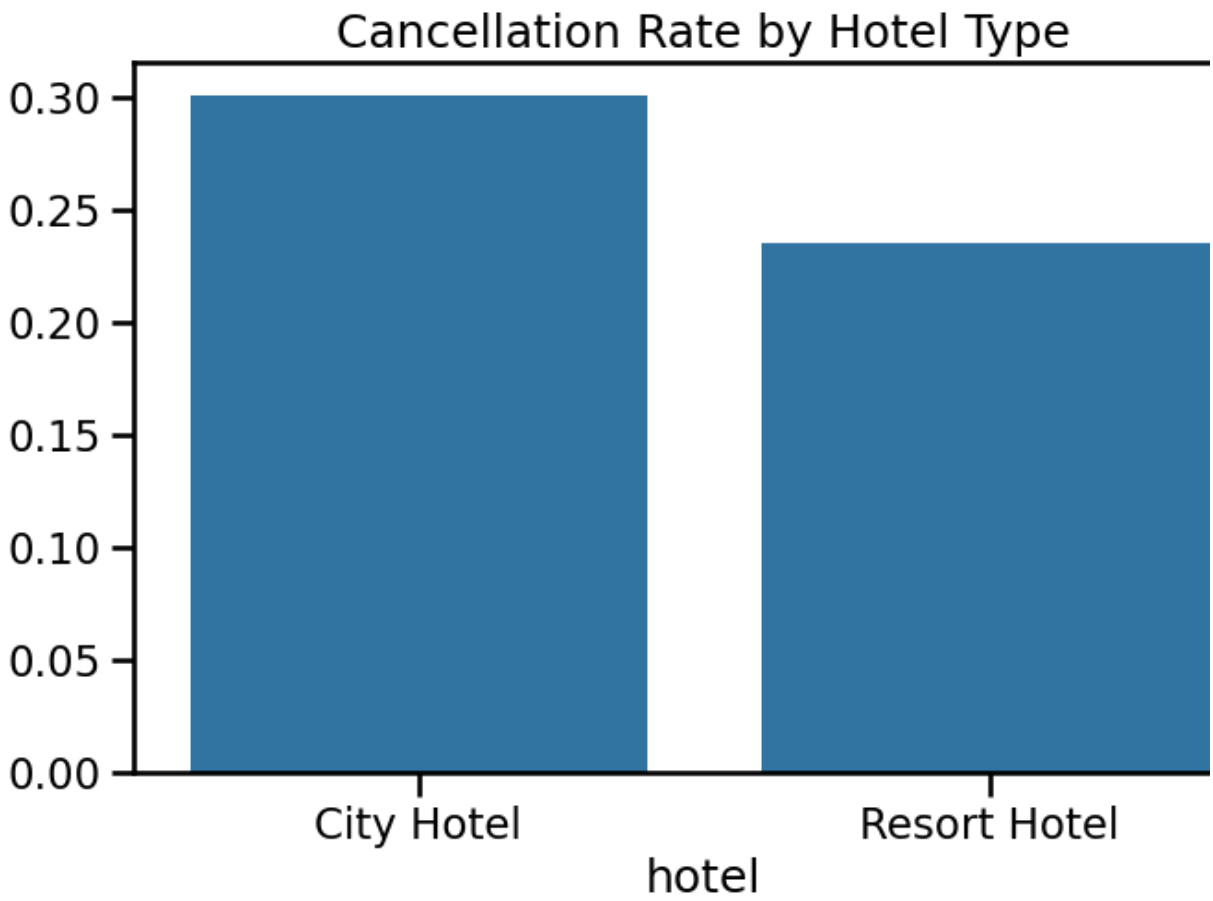
```
is_canceled
0    0.725102
1    0.274898
Name: proportion, dtype: float64
```

In [148…

```
# Calculate and visualize cancellation rate by hotel type.
cancel_rate_by_hotel = data.groupby('hotel')['is_canceled'].mean().sort_v
plt.figure(figsize=(8, 5))
sns.barplot(x=cancel_rate_by_hotel.index, y=cancel_rate_by_hotel.values)
plt.title('Cancellation Rate by Hotel Type')
```

Out[148…

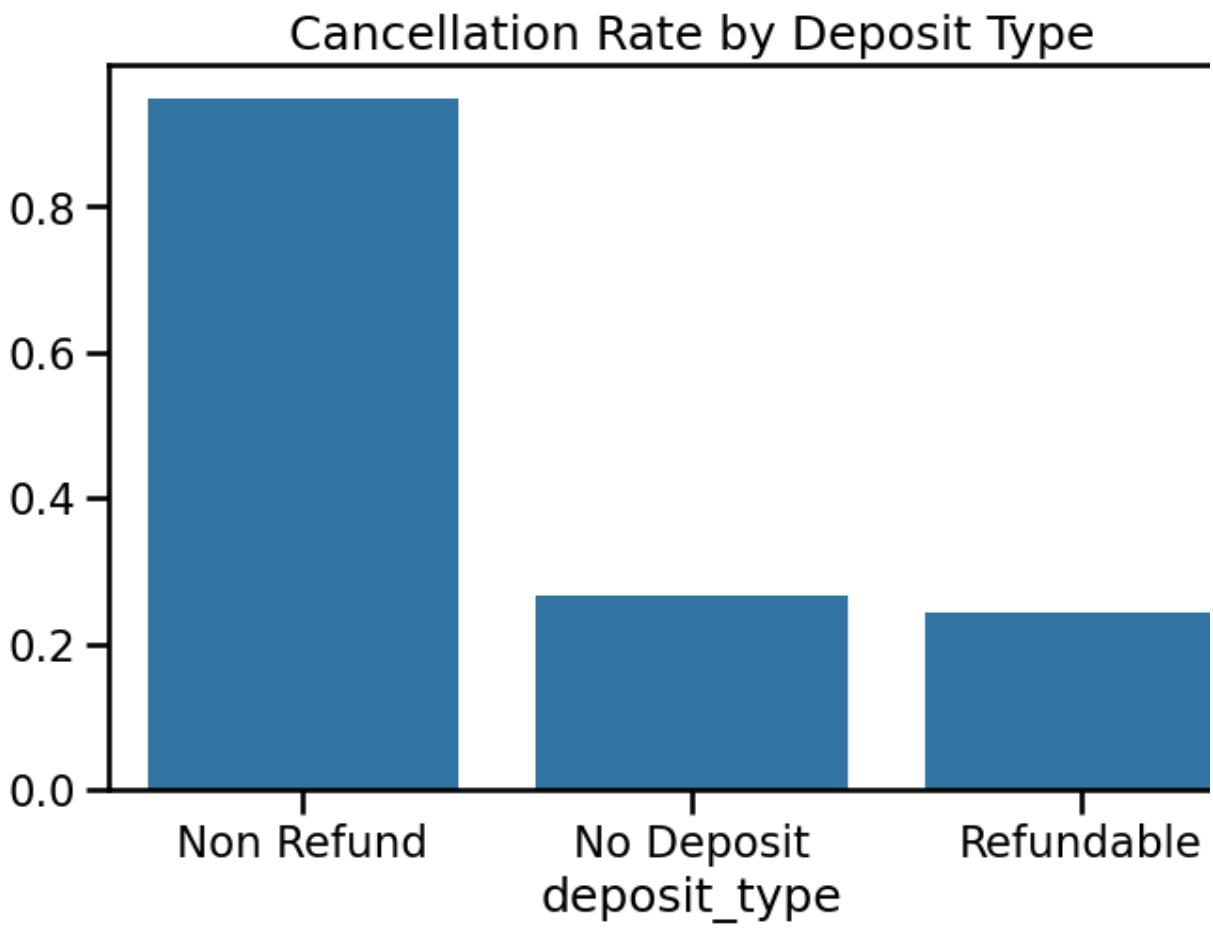
```
Text(0.5, 1.0, 'Cancellation Rate by Hotel Type')
```



In [149…

```
# Calculate and visualize cancellation rate by deposit type.
cancel_rate_by_deposit_type = data.groupby('deposit_type')['is_canceled']
plt.figure(figsize=(8, 5))
sns.barplot(x=cancel_rate_by_deposit_type.index, y=cancel_rate_by_deposit
plt.title('Cancellation Rate by Deposit Type')
```

Out[149… Text(0.5, 1.0, 'Cancellation Rate by Deposit Type')

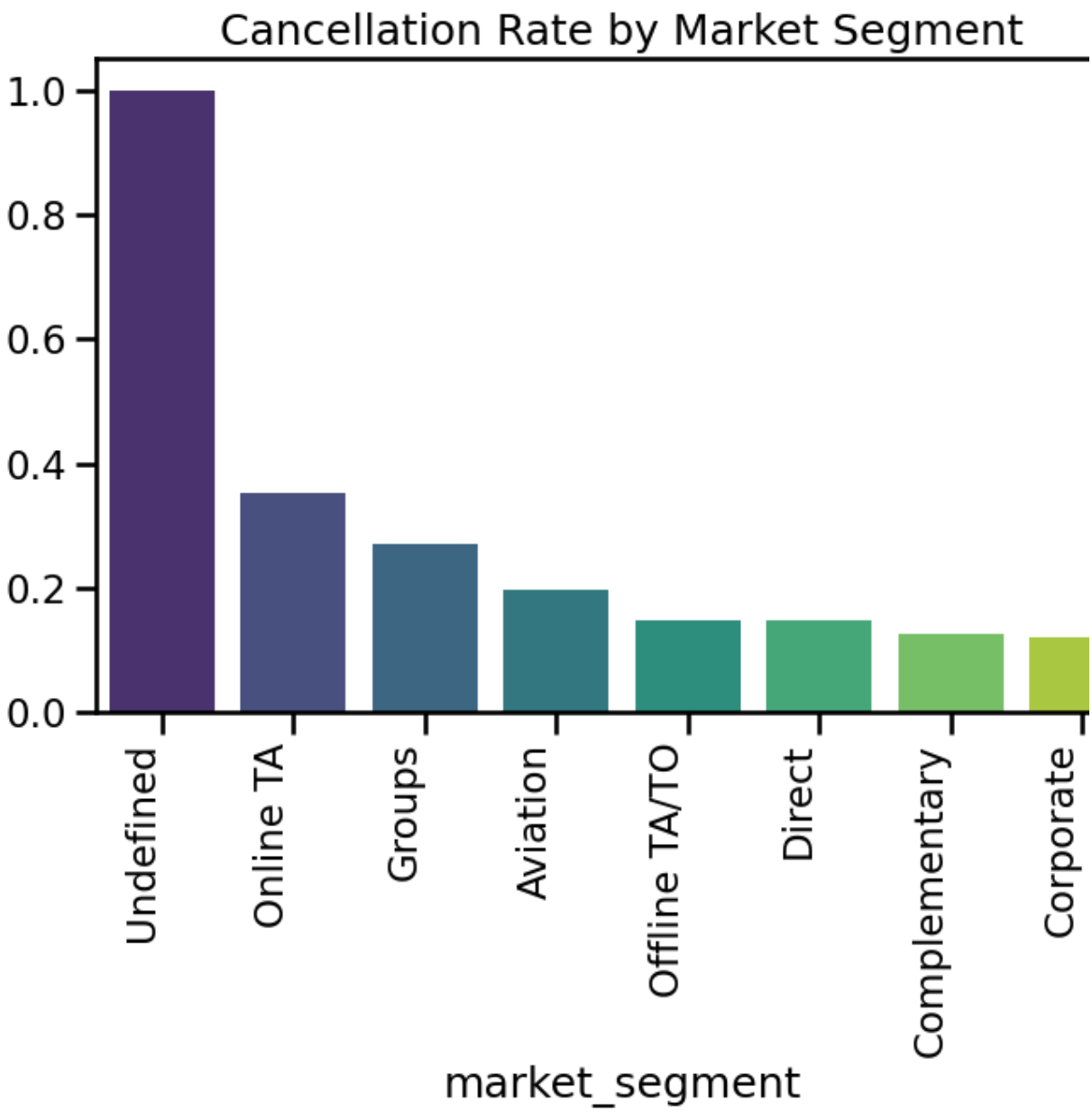


In [150…

```
# Calculate and visualize cancellation rate by market segment.
cancel_rate_by_market_segment = data.groupby('market_segment')['is_cancel
plt.figure(figsize=(8, 5))
sns.barplot(x=cancel_rate_by_market_segment.index, y=cancel_rate_by_marke
plt.xticks(rotation=90, ha='right')
plt.title('Cancellation Rate by Market Segment')
```

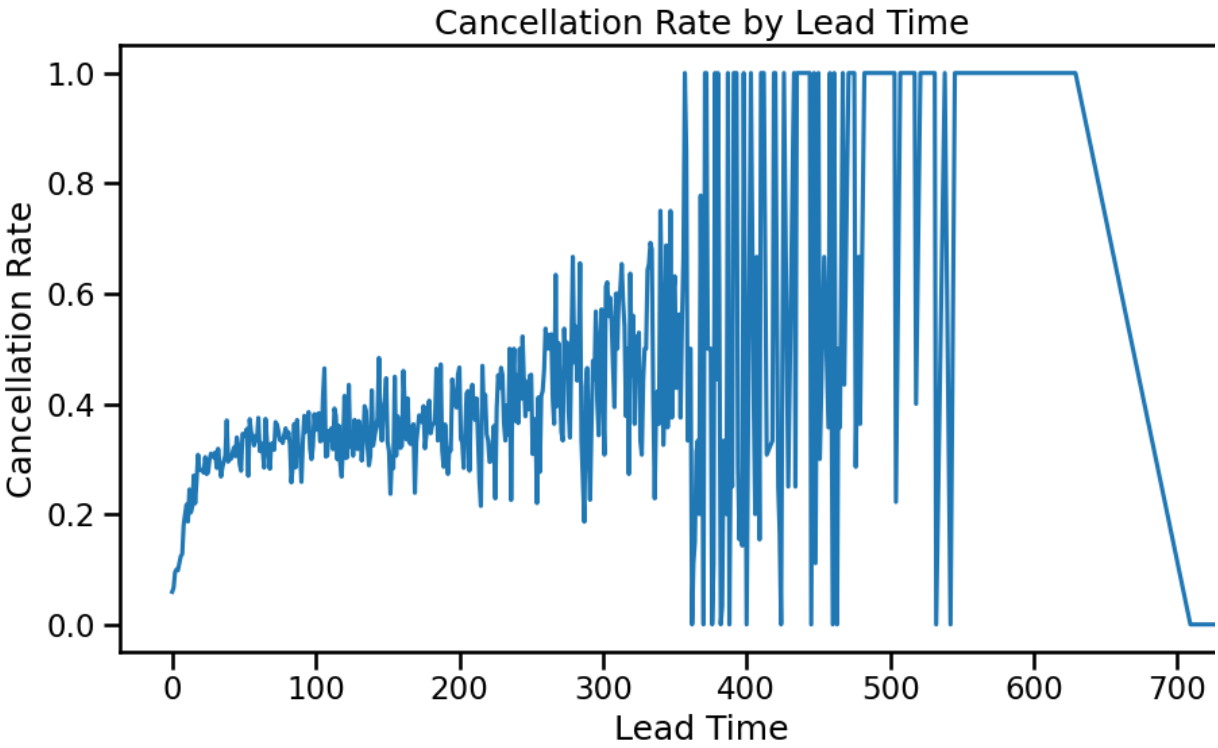
```
C:\Users\persh\AppData\Local\Temp\ipykernel_16684\3980639845.py:3: FutureWarn

Passing `palette` without assigning `hue` is deprecated and will be removed i
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.barplot(x=cancel_rate_by_market_segment.index,
y=cancel_rate_by_market_segment.values, palette='viridis')
```

Out[150… Text(0.5, 1.0, 'Cancellation Rate by Market Segment')



In [151…

```
# Visualize how cancellation rate changes with lead time.
cancel_rate_by_lead_time = data.groupby('lead_time')['is_canceled'].mean(
plt.figure(figsize=(10,6))
sns.lineplot(data=cancel_rate_by_lead_time.reset_index(), x='lead_time',
plt.title('Cancellation Rate by Lead Time')
plt.xlabel('Lead Time')
plt.ylabel('Cancellation Rate')
plt.tight_layout()
plt.show()
```



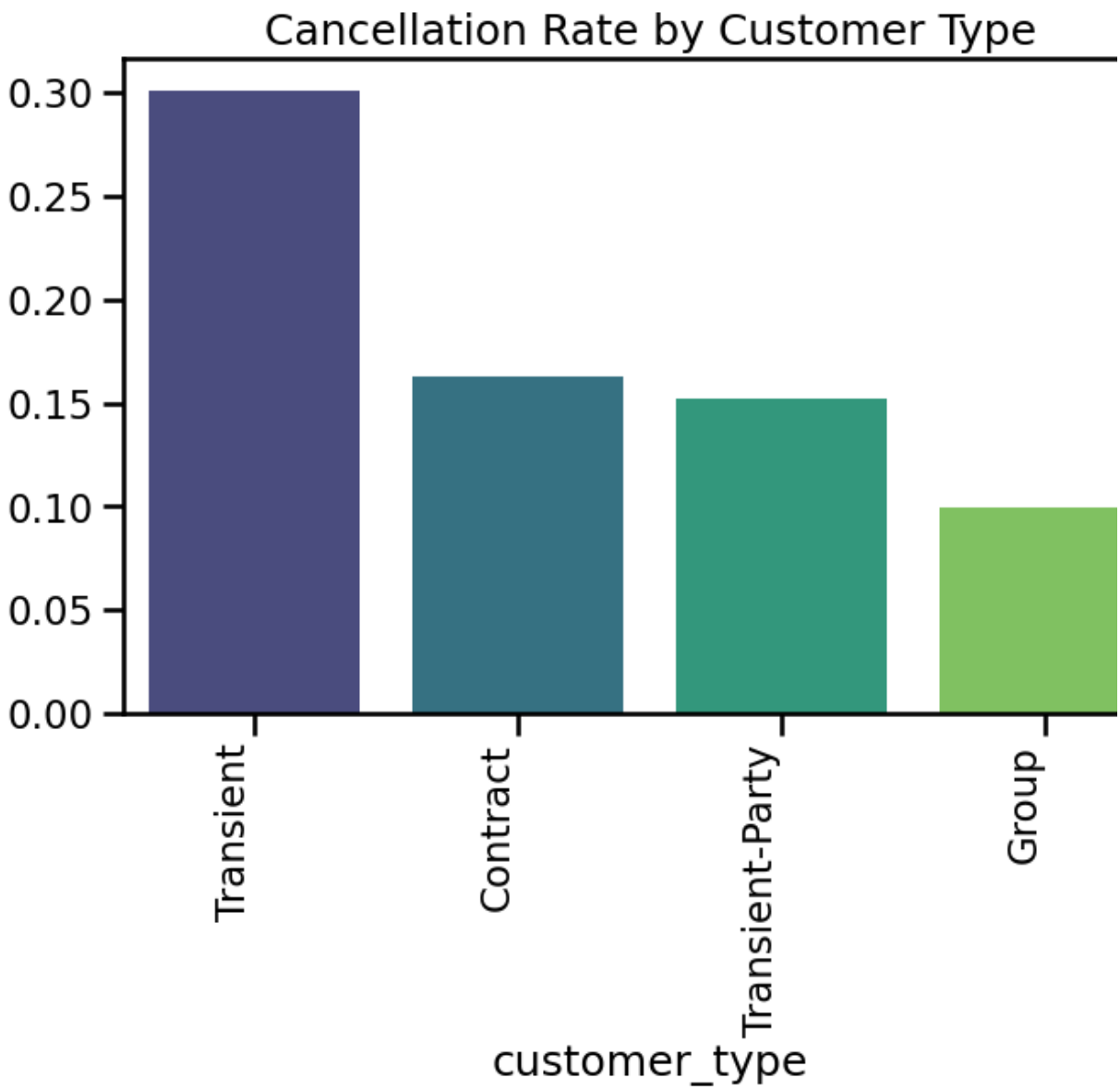
In [152…

```
# Calculate and visualize cancellation rate by customer type.
cancel_rate_by_customer_type = data.groupby('customer_type')['is_canceled
plt.figure(figsize=(8, 5))
sns.barplot(x=cancel_rate_by_customer_type.index, y=cancel_rate_by_custom
plt.xticks(rotation=90, ha='right')
plt.title('Cancellation Rate by Customer Type')
```

```
C:\Users\persh\AppData\Local\Temp\ipykernel_16684\228026964.py:3: FutureWarni

Passing `palette` without assigning `hue` is deprecated and will be removed i
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.barplot(x=cancel_rate_by_customer_type.index,
y=cancel_rate_by_customer_type.values, palette='viridis')
```

Out[152…

```
Text(0.5, 1.0, 'Cancellation Rate by Customer Type')
```



In [153…

```
# Display all column names to review available features.
data.columns
```

Out[153…

```
Index(['hotel', 'is_canceled', 'lead_time', 'arrival_date_year',
       'arrival_date_month', 'arrival_date_week_number',
       'arrival_date_day_of_month', 'stays_in_weekend_nights',
       'stays_in_week_nights', 'adults', 'children', 'babies', 'meal',
       'country', 'market_segment', 'distribution_channel',
       'is_repeated_guest', 'previous_cancellations',
       'previous_bookings_not_canceled', 'reserved_room_type',
       'assigned_room_type', 'booking_changes', 'deposit_type',
       'days_in_waiting_list', 'customer_type', 'adr',
       'required_car_parking_spaces', 'total_of_special_requests'],
      dtype='object')
```

In [154…

```
# Import preprocessing tools for scaling and encoding.
from sklearn.preprocessing import OneHotEncoder, MinMaxScaler, StandardSc
```

In [155…

```
# Take a random sample for faster pairplot visualization.
sample_data = data.sample(5000, random_state=42)
```

Quick interpretation:

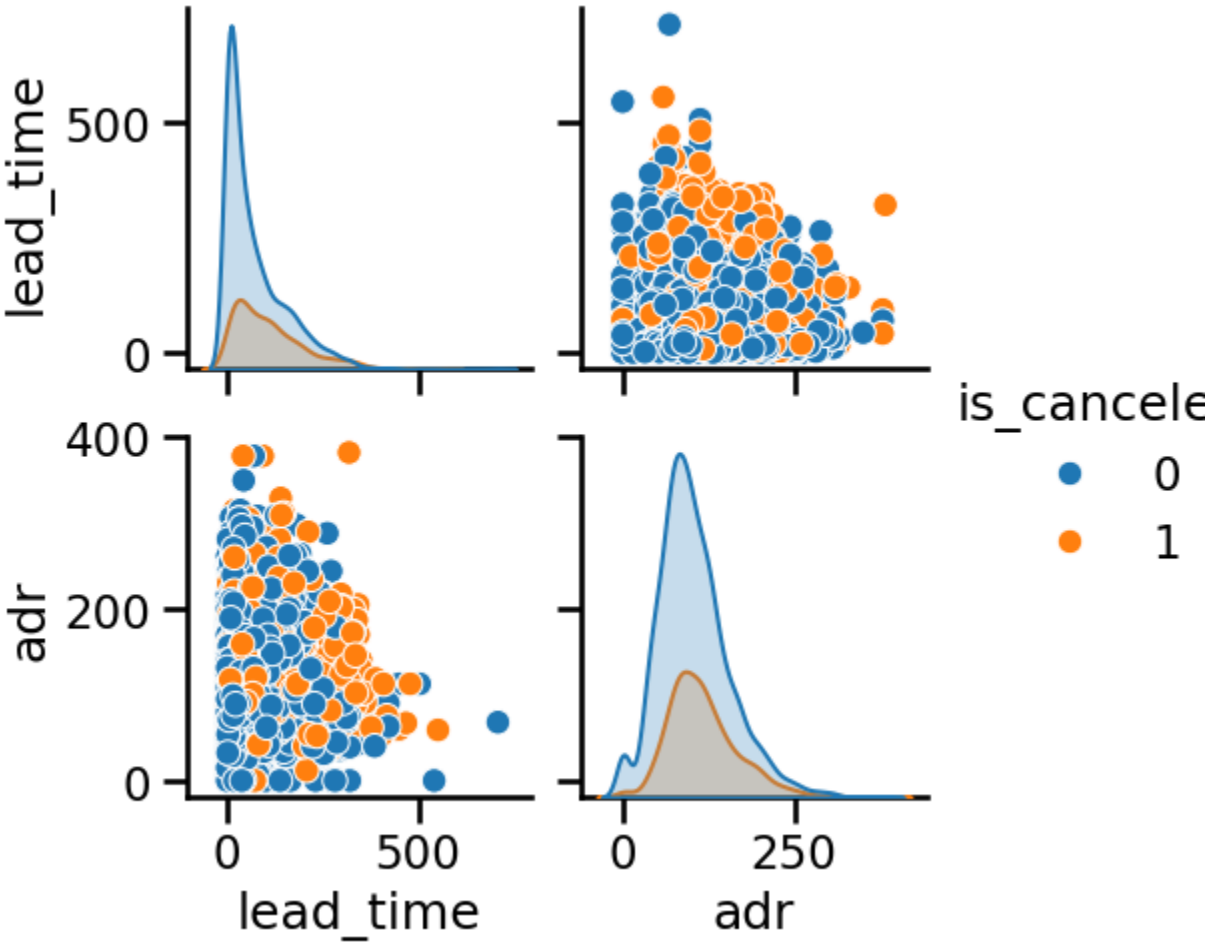
This pairplot compares `lead_time` and `adr` across canceled and non-canceled bookings. It helps ch whether canceled bookings tend to cluster around longer lead times or different price levels.

In [156…

```
# Create a pairplot to study the relationship between lead time, ADR, and
plot_data_1 = sample_data[['lead_time', 'adr', 'is_canceled']].copy()

g = sns.pairplot(plot_data_1, hue='is_canceled', corner=True, diag_kind='

for ax in g.axes.flatten():
    if ax is not None:
        ax.tick_params(axis='x', labelrotation=35)
        ax.tick_params(axis='y', labelrotation=0)
        ax.set_xlabel(ax.get_xlabel(), fontsize=9)
        ax.set_ylabel(ax.get_ylabel(), fontsize=9)

g.fig.subplots_adjust(top=0.92, wspace=0.15, hspace=0.15)
g.fig.suptitle('Pairplot of Lead Time and ADR by Cancellation Status', fo

plt.show()
```

Out[156…

```
<seaborn.axisgrid.PairGrid at 0x29b7a8dc050>
```



Quick interpretation:

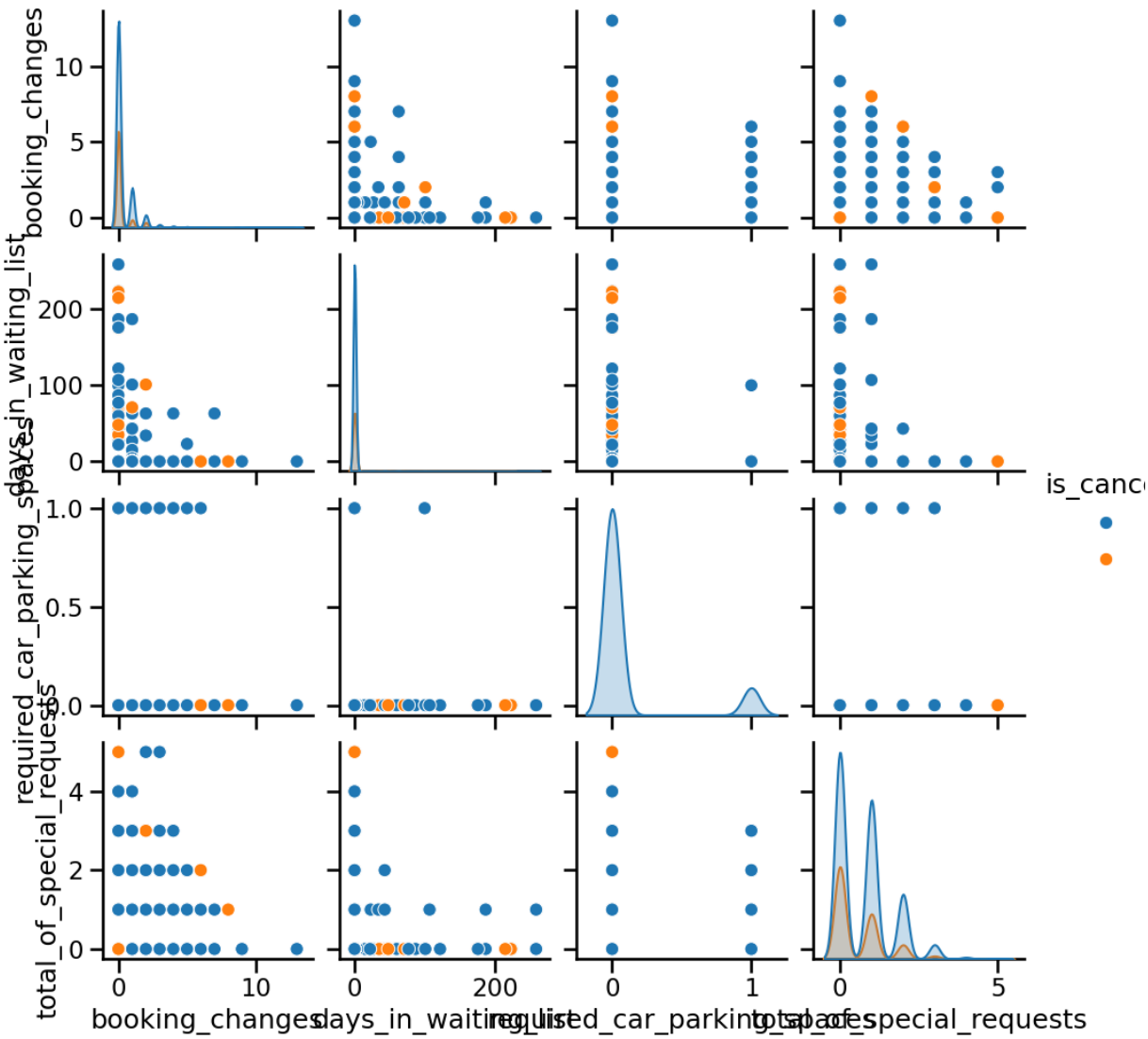
This pairplot focuses on booking behavior variables such as booking changes, waiting-list days, parking de

special requests. It is useful for visually checking whether customer behavior patterns differ between cance

non-canceled bookings.

In [ ]:
```
# Create a second pairplot with shorter axis labels so the figure is easi
plot_data_2 = sample_data[[
    'booking_changes',
    'days_in_waiting_list',
    'required_car_parking_spaces',
    'total_of_special_requests',
    'is_canceled'
]].rename(columns={
    'booking_changes': 'booking_changes',
    'days_in_waiting_list': 'waiting_list_days',
    'required_car_parking_spaces': 'parking_spaces',
    'total_of_special_requests': 'special_requests'
}).copy()

g = sns.pairplot(plot_data_2, hue='is_canceled', corner=True, diag_kind='

for ax in g.axes.flatten():
    if ax is not None:
        ax.tick_params(axis='x', labelrotation=35)
        ax.tick_params(axis='y', labelrotation=0)
        ax.set_xlabel(ax.get_xlabel(), fontsize=9)
        ax.set_ylabel(ax.get_ylabel(), fontsize=9)

g.fig.subplots_adjust(top=0.93, wspace=0.15, hspace=0.15)
g.fig.suptitle('Pairplot of Booking Behavior Features by Cancellation Sta

plt.show()
```

Out[ ]:
```
<seaborn.axisgrid.PairGrid at 0x29b60877e10>
```



In [158…

```
# Convert selected categorical variables into dummy variables for modelin
categorical_cols = [
    'hotel',
    'meal',
    'market_segment',
    'distribution_channel',
    'deposit_type',
    'customer_type',
    'reserved_room_type',
    'assigned_room_type',
    'arrival_date_month',
    'country'
]

data = pd.get_dummies(data, columns=categorical_cols, drop_first=True)
```

In [159…

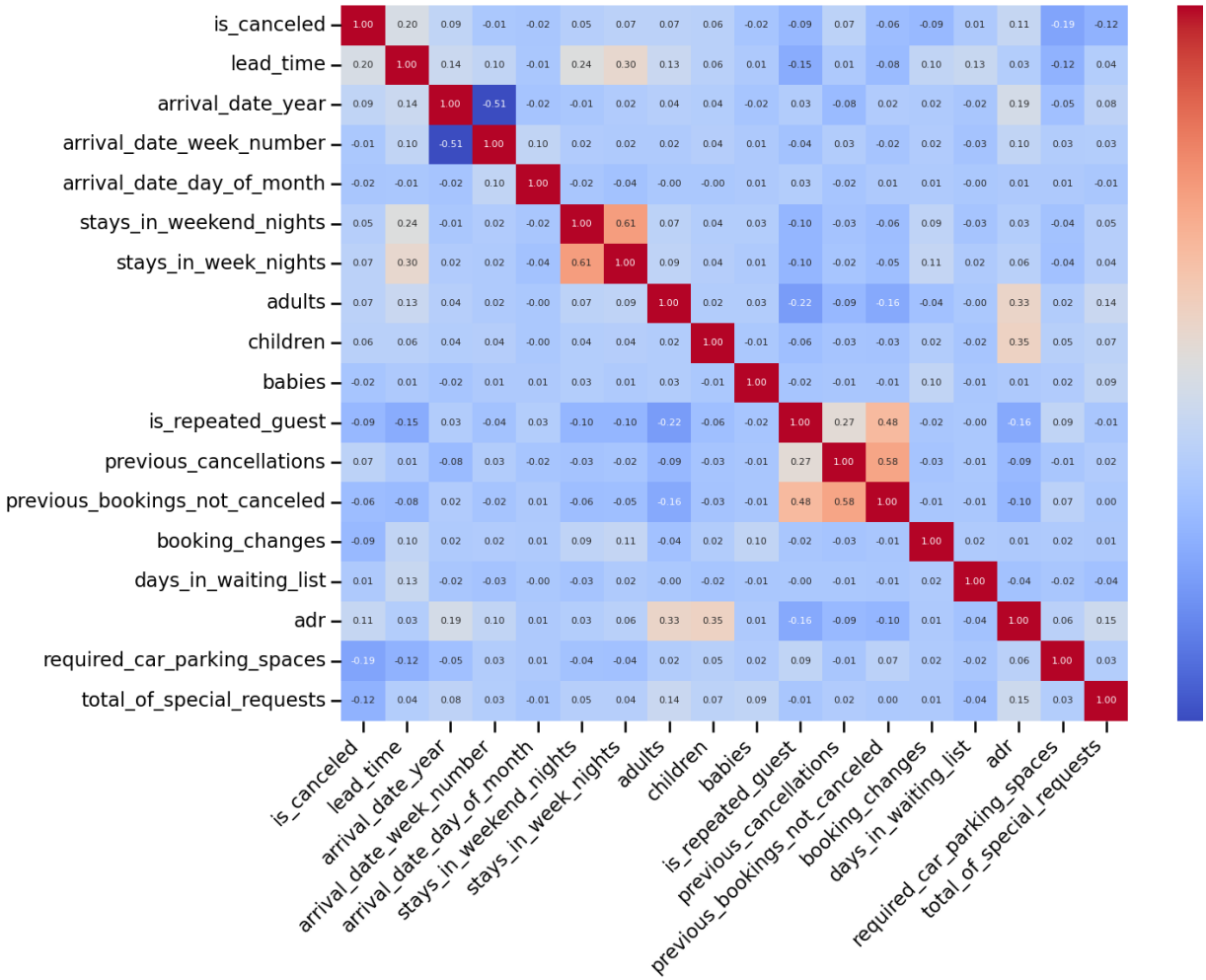
```
# Check the dataset shape after one-hot encoding.
data.shape
```

Out[159… (87396, 246)

In [160…

```
# Plot a correlation heatmap for the numeric variables in the sample.
plt.figure(figsize=(16,12))
corr = sample_data.corr(numeric_only=True)
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', annot_kws={'siz
plt.xticks(rotation=45, ha='right')
```

```
plt.yticks(rotation=0)
plt.tight_layout()
```

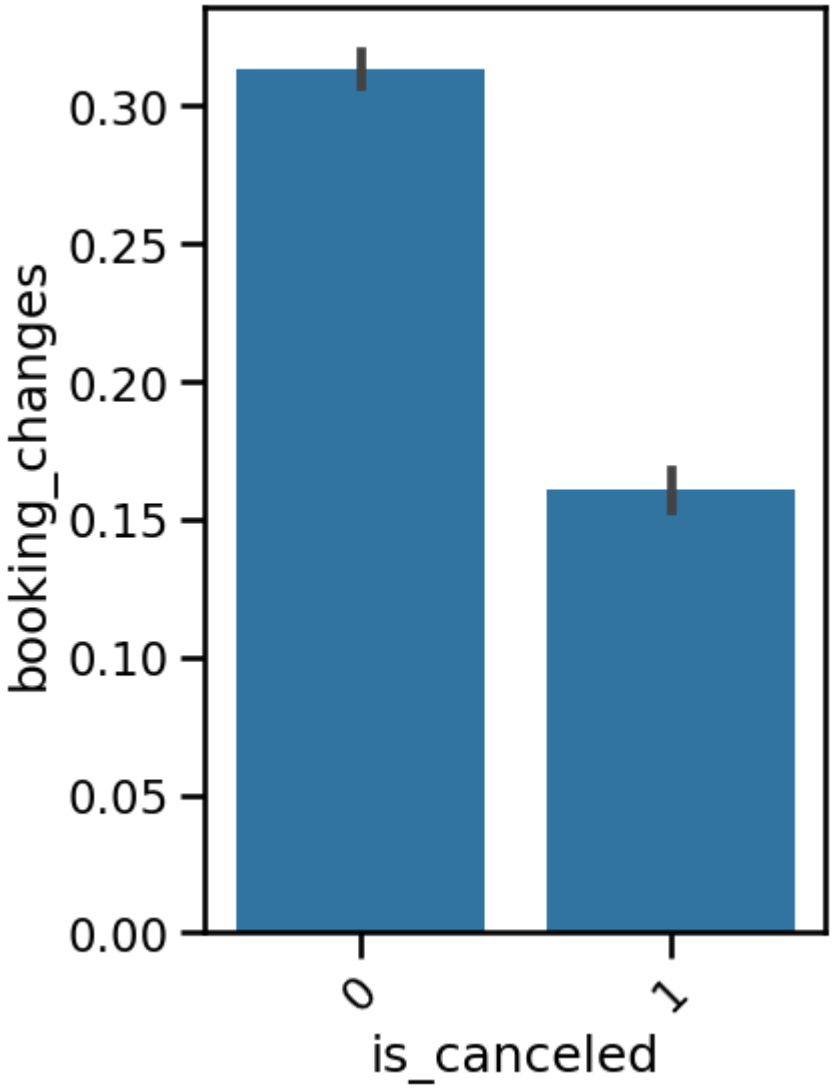


In [161…

```
# Compare average booking changes for canceled vs. non-canceled bookings.
plt.figure(figsize=(4,6))
sns.barplot(x = 'is_canceled', y='booking_changes', data=data)
plt.xticks(rotation=45)
```

Out[161…

```
([0, 1], [Text(0, 0, '0'), Text(1, 0, '1')])
```

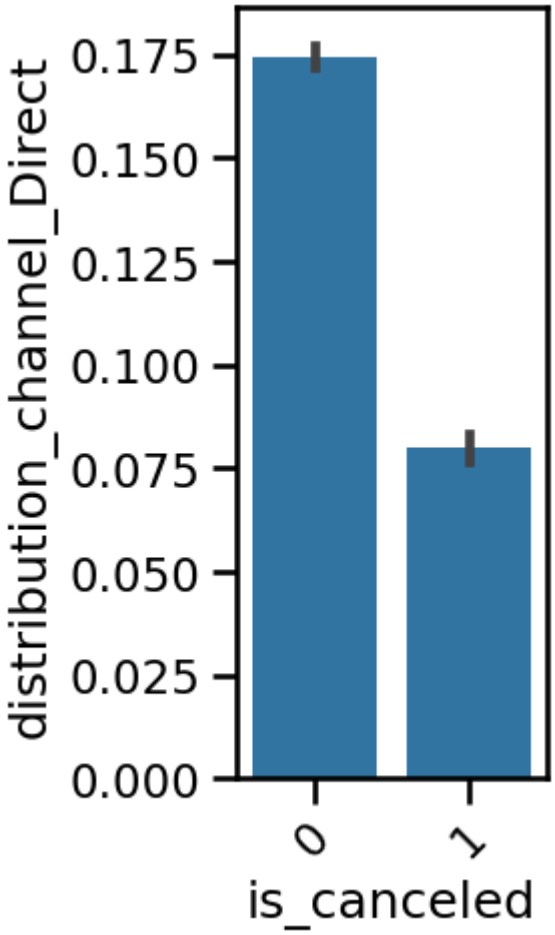


In [200…

```
# Check whether direct distribution channel is related to cancellation st
plt.figure(figsize=(2,5))
sns.barplot(x = 'is_canceled', y='distribution_channel_Direct', data=data
plt.xticks(rotation=45)
```

Out[200…

```
([0, 1], [Text(0, 0, '0'), Text(1, 0, '1')])
```



In [164…

```
# Display summary statistics for the cleaned dataset.
data.describe().T
```

Out[164…

	count	mean
is_canceled	87396.0	0.274898	0.44646
lead_time	87396.0	79.891368	86.05232
arrival_date_year	87396.0	2016.210296	0.68610
arrival_date_week_number	87396.0	26.838334	13.67457
arrival_date_day_of_month	87396.0	15.815541	8.83514
stays_in_weekend_nights	87396.0	1.005263	1.03192
stays_in_week_nights	87396.0	2.625395	2.05358
adults	87396.0	1.875795	0.62650
children	87396.0	0.138633	0.45587
babies	87396.0	0.010824	0.11359
is_repeated_guest	87396.0	0.039075	0.19377
previous_cancellations	87396.0	0.030413	0.36914
previous_bookings_not_canceled	87396.0	0.183990	1.73189
booking_changes	87396.0	0.271603	0.72724
days_in_waiting_list	87396.0	0.749565	10.01573
adr	87396.0	106.337246	55.01395
required_car_parking_spaces	87396.0	0.084226	0.28153
total_of_special_requests	87396.0	0.698567	0.83194

In [165…

```
# Detect outliers in 'lead_time' using the IQR method.
Q1= data['lead_time'].quantile(0.25)
Q3= data['lead_time'].quantile(0.75)
IQR= Q3 - Q1
lower_band = Q1- 1.5 * IQR
upper_band = Q3 + 1.5 * IQR

outliers = data[(data['lead_time'] < lower_band) | (data['lead_time'] > u
print(outliers)
```

```
        is_canceled  lead_time  arrival_date_year  arrival_date_week_number
0                 0        342               2015                        27
1                 0        737               2015                        27
81                1        368               2015                        27
82                0        364               2015                        27
85                0        324               2015                        27
...             ...        ...                ...                       ...
119148            0        457               2017                        34
119228            0        341               2017                        35
119296            0        303               2017                        35
119297            0        332               2017                        35
119334            0        325               2017                        35

        arrival_date_day_of_month  stays_in_weekend_nights  \
0                               1                        0
1                               1                        0
81                              3                        3
82                              3                        3
85                              3                        4
...                           ...                      ...
119148                         25                        2
119228                         27                        2
119296                         29                        0
119297                         31                        0
119334                         30                        0

        stays_in_week_nights  adults  children  babies  ...  country_UMI  \
0                          0       2       0.0       0  ...        False
1                          0       2       0.0       0  ...        False
81                         7       2       0.0       0  ...        False
82                         7       2       0.0       0  ...        False
85                        10       2       0.0       0  ...        False
...                      ...     ...       ...     ...  ...          ...
119148                     3       2       0.0       0  ...        False
119228                     3       2       1.0       0  ...        False
119296                     4       1       0.0       0  ...        False
119297                     2       2       0.0       0  ...        False
119334                     4       2       0.0       0  ...        False

        country_URY  country_USA  country_UZB  country_VEN  country_VGB  \
0             False        False        False        False        False
1             False        False        False        False        False
81            False        False        False        False        False
82            False        False        False        False        False
85            False        False        False        False        False
...             ...          ...          ...          ...          ...
119148        False        False        False        False        False
119228        False        False        False        False        False
119296        False        False        False        False        False
119297        False        False        False        False        False
119334        False        False        False        False        False

        country_VNM  country_ZAF  country_ZMB  country_ZWE
0             False        False        False        False
1             False        False        False        False
81            False        False        False        False
82            False        False        False        False
85            False        False        False        False
...             ...          ...          ...          ...
119148        False        False        False        False
```

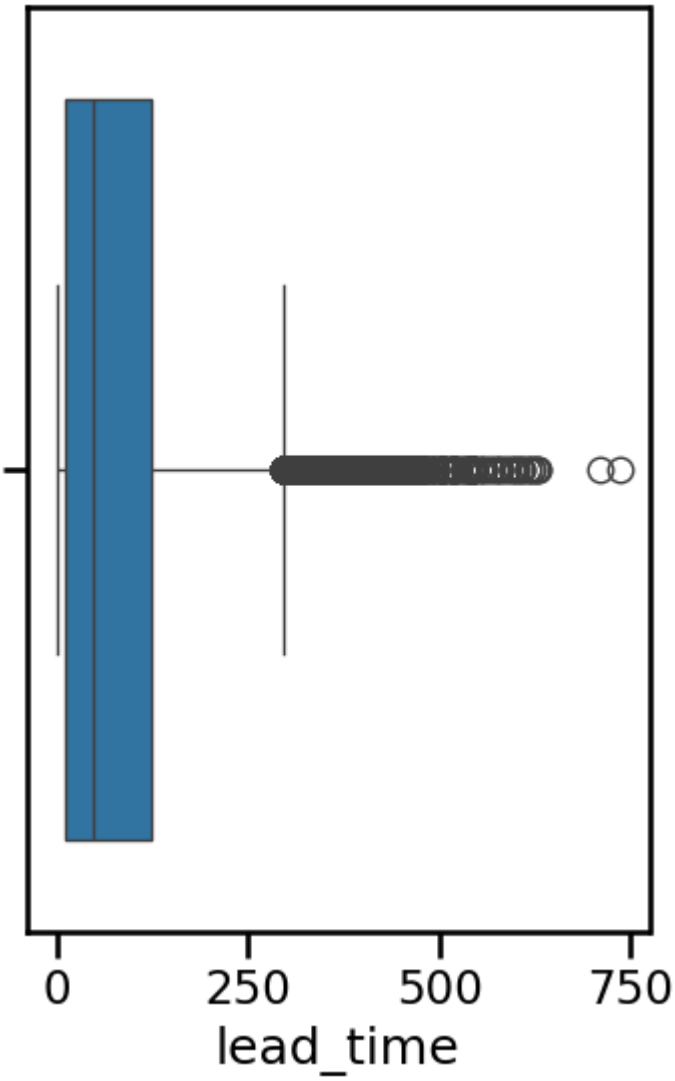
```
119228        False        False        False        False
119296        False        False        False        False
119297        False        False        False        False
119334        False        False        False        False

[2396 rows x 246 columns]
```

In [166…

```
# Visualize the distribution and outliers of lead time with a boxplot.
plt.figure(figsize=(4,6))
sns.boxplot(x=data['lead_time'])
```

Out[166…

```
<Axes: xlabel='lead_time'>
```



In [167…

```
# Define a helper function to compute the main classification metrics.
from sklearn.metrics import accuracy_score, precision_score, recall_score

def get_metrics(y_test, y_pred, y_pred_proba=None):
    results = {}
    results['accuracy'] = accuracy_score(y_test, y_pred)
    results['precision'] = precision_score(y_test, y_pred)
    results['recall'] = recall_score(y_test, y_pred)
    results['f1_score'] = f1_score(y_test, y_pred)

    if y_pred_proba is not None:
        results['roc_auc'] = roc_auc_score(y_test, y_pred_proba)

    print("Confusion Matrix:")
```

```
    print(confusion_matrix(y_test, y_pred))

    print("\nClassification Report:")
    print(classification_report(y_test, y_pred))

    return results
```

In [168…

```
# Define helper functions to extract and visualize model coefficients.
# Extract and sort feature coefficients
def get_feature_coefs(regression_model, label_index, columns):
    coef_dict = {}
    for coef, feat in zip(regression_model.coef_[label_index, :], columns
        if abs(coef) >= 0.01:
            coef_dict[feat] = coef
    # Sort coefficients
    coef_dict = {k: v for k, v in sorted(coef_dict.items(), key=lambda it
    return coef_dict

# Generate bar colors based on if value is negative or positive
def get_bar_colors(values):
    color_vals = []
    for val in values:
        if val <= 0:
            color_vals.append('r')
        else:
            color_vals.append('g')
    return color_vals

# Visualize coefficients
def visualize_coefs(coef_dict):
    features = list(coef_dict.keys())
    values = list(coef_dict.values())
    y_pos = np.arange(len(features))
    color_vals = get_bar_colors(values)
    plt.rcdefaults()
    fig, ax = plt.subplots()
    ax.barh(y_pos, values, align='center', color=color_vals)
    ax.set_yticks(y_pos)
    ax.set_yticklabels(features)
    # labels read top-to-bottom
    ax.invert_yaxis()
    ax.set_xlabel('Feature Coefficients')
    ax.set_title('')
    plt.show()
```

In [169…

```
# Separate features (X) from the target variable (y).
X = data.loc[: , data.columns!='is_canceled']
y=data.loc[:, 'is_canceled']
```

In [170…

```
# Split the dataset into training and testing sets using stratification.
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
```

In [171…

```
# Standardize the features for models that are sensitive to scale.
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Logistic Regression

In [172…

```
# Train the Logistic Regression model on the scaled training data.
from sklearn.linear_model import LogisticRegression

LR_model = LogisticRegression(max_iter=1000, random_state=42)
LR_model.fit(X_train_scaled, y_train)
```

Out[172…

LogisticRegression

Parameters

penalty	'l2'
dual	False
tol	0.0001
C	1.0
fit_intercept	True
intercept_scaling	1
class_weight	None
random_state	42
solver	'lbfgs'
max_iter	1000
multi_class	'deprecated'
verbose	0
warm_start	False
n_jobs	None
l1_ratio	None

In [173…

```
# Make predictions and evaluate Logistic Regression.
y_pred = LR_model.predict(X_test_scaled)
y_pred_proba = LR_model.predict_proba(X_test_scaled)[:, 1]

scores_LR = get_metrics(y_test, y_pred, y_pred_proba)
print(scores_LR)
```

```
Confusion Matrix:
[[11558  1117]
 [ 2482  2323]]

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.91      0.87     12675
           1       0.68      0.48      0.56      4805

    accuracy                           0.79     17480
   macro avg       0.75      0.70      0.71     17480
weighted avg       0.78      0.79      0.78     17480

{'accuracy': 0.7941075514874142, 'precision': 0.6752906976744186, 'recall':
0.48345473465140476, 'f1_score': 0.5634930260764099, 'roc_auc': 0.84379224468
```

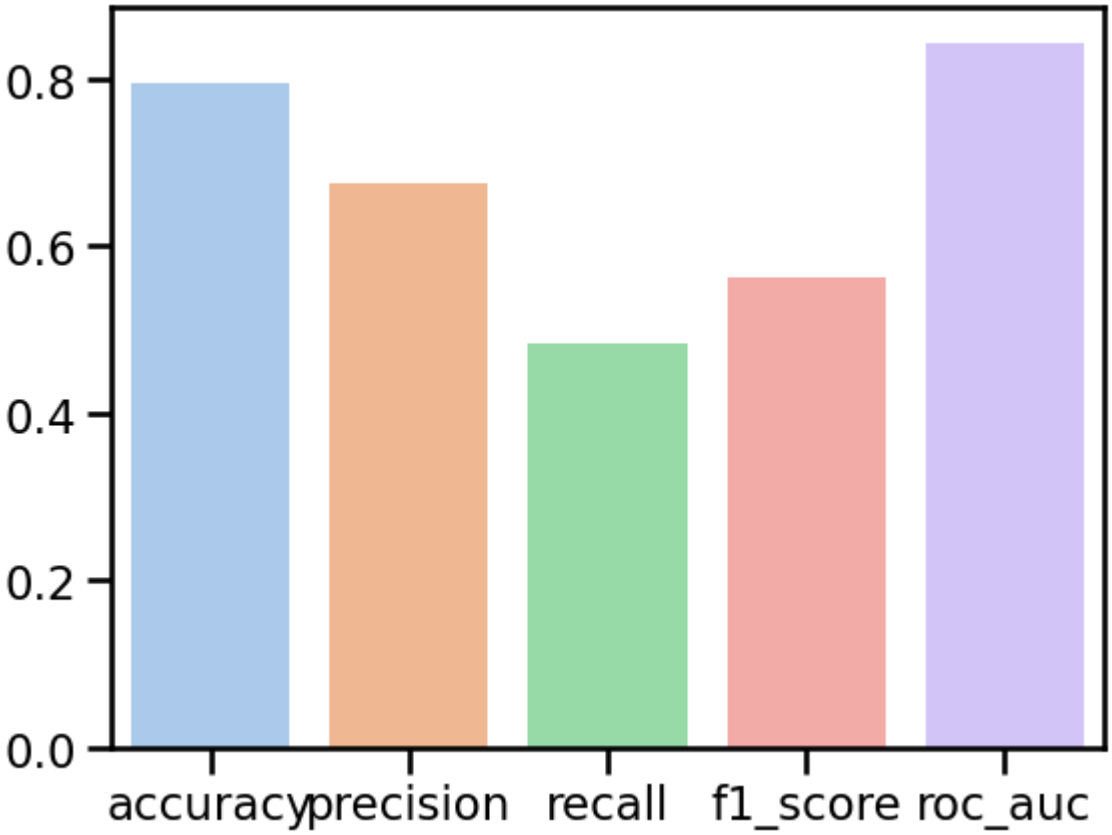
In [174…

```python
# Plot the main performance metrics for Logistic Regression.
plt.figure(figsize=(8, 4))
sns.barplot(x=list(scores_LR.keys()), y=list(scores_LR.values()), hue=lis
plt.title('Logistic Regression Performance Metrics')
plt.xlabel('Metric')
plt.ylabel('Score')
plt.xticks(rotation=30)
plt.ylim(0, 1)
plt.tight_layout()
plt.show()
```

```
C:\Users\persh\AppData\Local\Temp\ipykernel_16684\2439316481.py:1: FutureWarn

Passing `palette` without assigning `hue` is deprecated and will be removed i
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.barplot(scores_LR, palette= 'pastel')
```

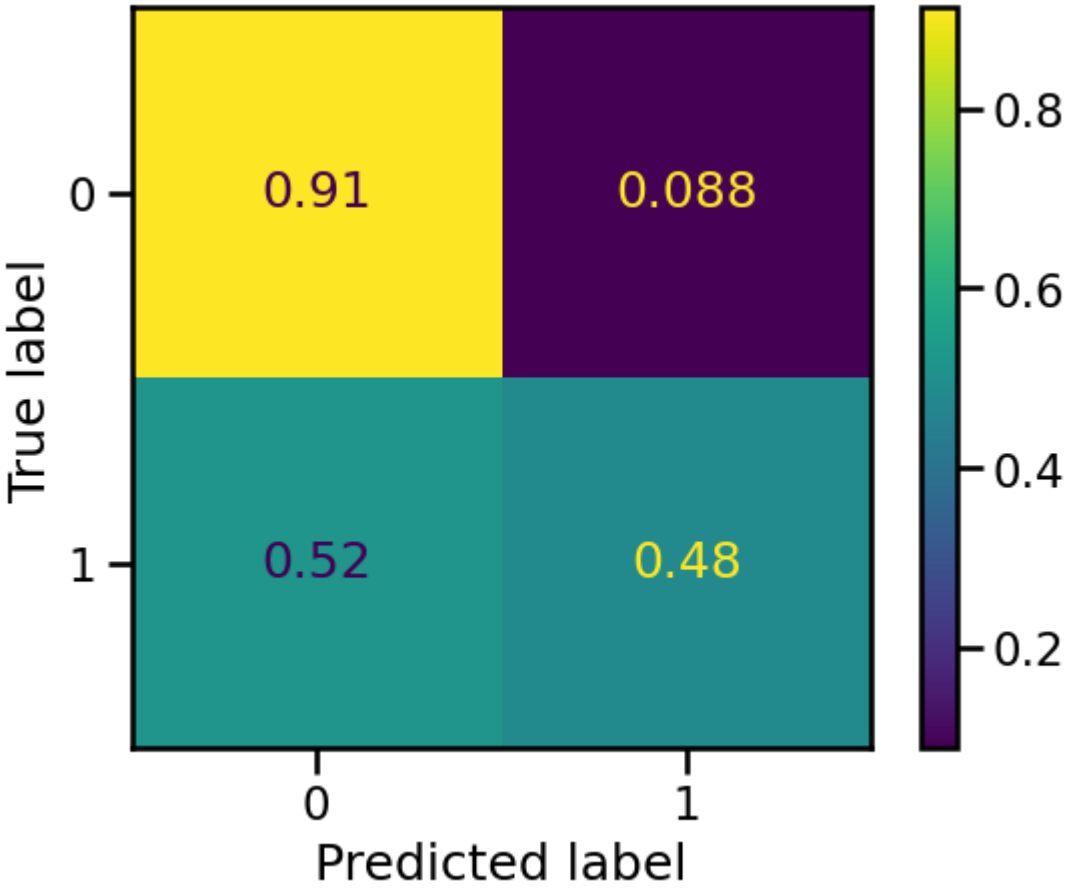
Out[174…

```
<Axes: >
```



In [175…

```
# Plot the normalized confusion matrix for Logistic Regression.
from sklearn.metrics import ConfusionMatrixDisplay
cf = confusion_matrix(y_test, y_pred, normalize='true')


sns.set_context('talk')
disp = ConfusionMatrixDisplay(confusion_matrix=cf, display_labels=LR_mode
disp.plot()
plt.show()
```



In [176…

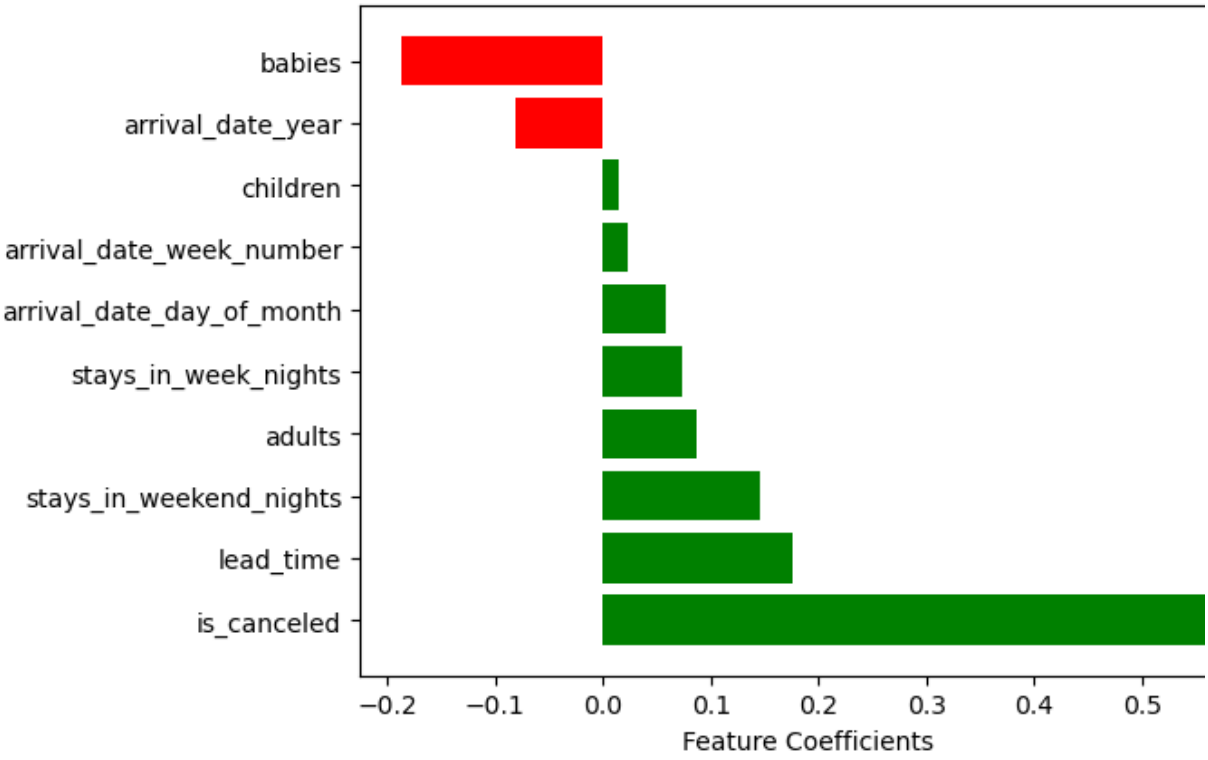
```
# Review the encoded feature names before coefficient interpretation.
data.columns.T
```

Out[176…

```
Index(['is_canceled', 'lead_time', 'arrival_date_year',
       'arrival_date_week_number', 'arrival_date_day_of_month',
       'stays_in_weekend_nights', 'stays_in_week_nights', 'adults', 'childre
       'babies',
       ...
       'country_UMI', 'country_URY', 'country_USA', 'country_UZB',
       'country_VEN', 'country_VGB', 'country_VNM', 'country_ZAF',
       'country_ZMB', 'country_ZWE'],
      dtype='object', length=246)
```

In [177…

```
# Visualize selected Logistic Regression coefficients.
coef_dict = get_feature_coefs(LR_model,label_index = 0, columns=['is_canc
       'arrival_date_week_number', 'arrival_date_day_of_month',
       'stays_in_weekend_nights', 'stays_in_week_nights', 'adults', 'chil
       'babies'])

visualize_coefs(coef_dict)
```



In [178…

```
# Train a Linear SVC model on the scaled data.
from sklearn.svm import LinearSVC

LinearSVC_model = LinearSVC(random_state=42, max_iter=10000)
LinearSVC_model.fit(X_train_scaled, y_train)
```

Out[178…

LinearSVC

Parameters

penalty	'l2'
loss	'squared_hinge'
dual	'auto'
tol	0.0001
C	1.0
multi_class	'ovr'
fit_intercept	True
intercept_scaling	1
class_weight	None
verbose	0
random_state	42
max_iter	10000

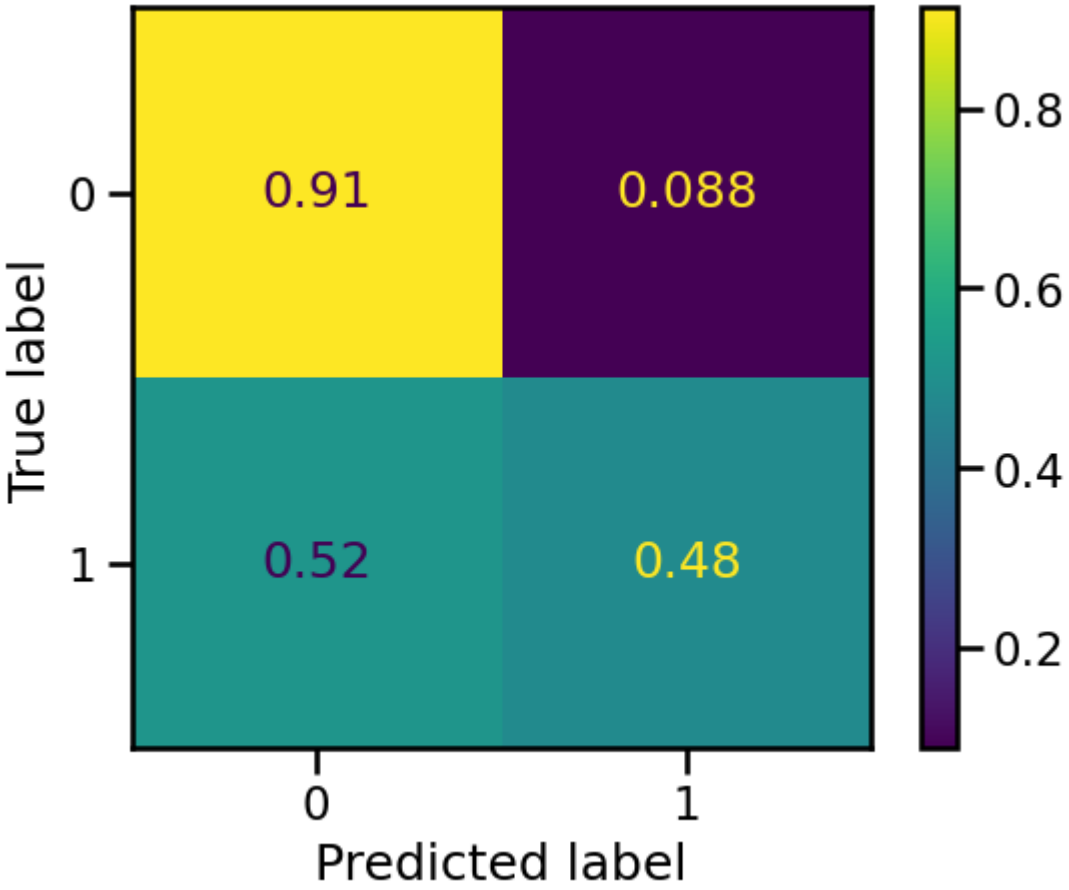
In [179…

```
# Evaluate the Linear SVC model.
y_pred = LinearSVC_model.predict(X_test_scaled)

scores_SVC = get_metrics(y_test, y_pred)
print(scores_SVC)
```

```
Confusion Matrix:
[[11662  1013]
 [ 2591  2214]]

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.92      0.87     12675
           1       0.69      0.46      0.55      4805

    accuracy                           0.79     17480
   macro avg       0.75      0.69      0.71     17480
weighted avg       0.78      0.79      0.78     17480

{'accuracy': 0.7938215102974828, 'precision': 0.6860861481251936, 'recall':
0.4607700312174818, 'f1_score': 0.5512948207171314}
```

In [180…

```
# Plot the normalized confusion matrix for Linear SVC.
cf_SVC = confusion_matrix(y_test, y_pred, normalize='true')

sns.set_context('talk')
disp = ConfusionMatrixDisplay(confusion_matrix=cf_SVC, display_labels=Lin
disp.plot()
plt.show()
```



KNN Classifier

In [181…

```
# Train the default K-Nearest Neighbors model.
from sklearn.neighbors import KNeighborsClassifier

KNN_model = KNeighborsClassifier()

KNN_model.fit(X_train_scaled, y_train)


y_pred = KNN_model.predict(X_test_scaled)
```

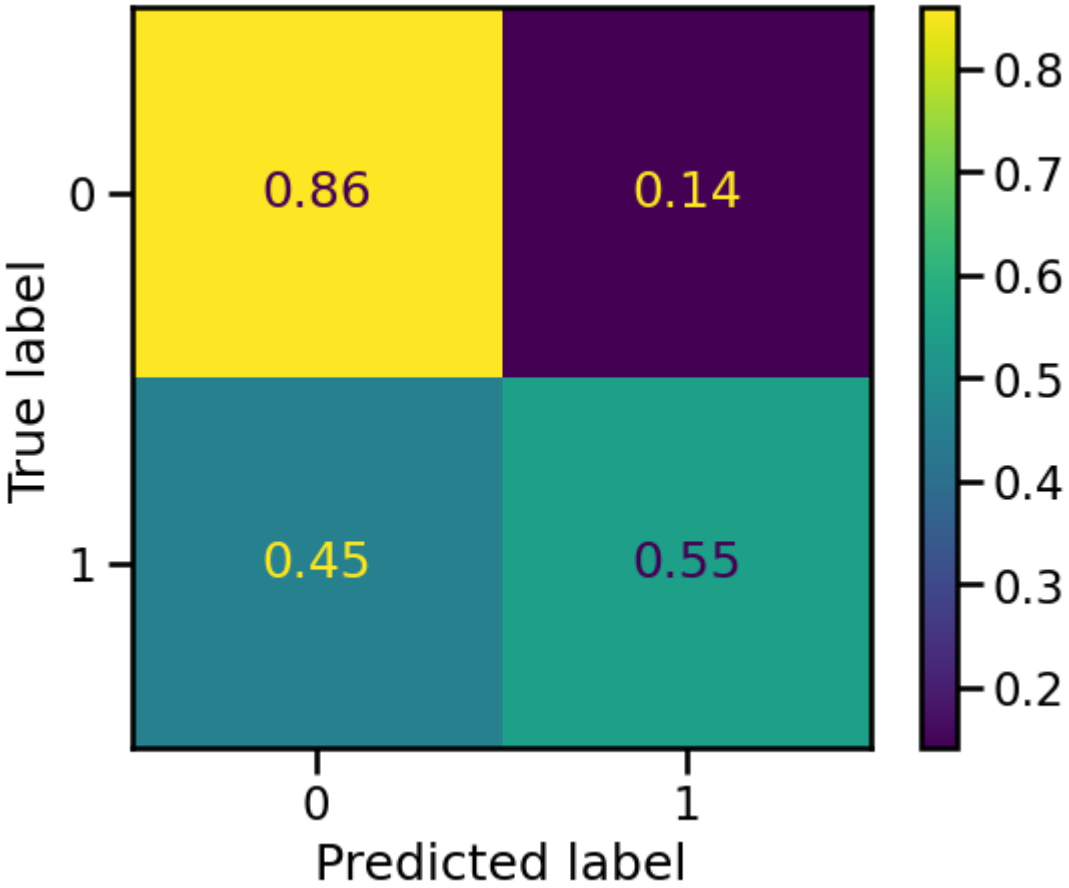
In [182…

```
# Evaluate the default KNN model.
scores_KNN = get_metrics(y_test, y_pred)
print(scores_KNN)
```

```
Confusion Matrix:
[[10888  1787]
 [ 2185  2620]]

Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.86      0.85     12675
           1       0.59      0.55      0.57      4805

    accuracy                           0.77     17480
   macro avg       0.71      0.70      0.71     17480
weighted avg       0.77      0.77      0.77     17480

{'accuracy': 0.7727688787185355, 'precision': 0.5945087361016564, 'recall':
0.5452653485952134, 'f1_score': 0.5688232739904473}
```

In [183…

```
# Plot the normalized confusion matrix for the default KNN model.
cf_KNN = confusion_matrix(y_test, y_pred, normalize='true')


sns.set_context('talk')
disp = ConfusionMatrixDisplay(confusion_matrix=cf_KNN, display_labels=KNN
disp.plot()
plt.show()
```



In [184…

```
# Define a tuned KNN model with distance weighting.
KNN_model_q = KNeighborsClassifier(n_neighbors=5, weights='distance', met
```

In [185…

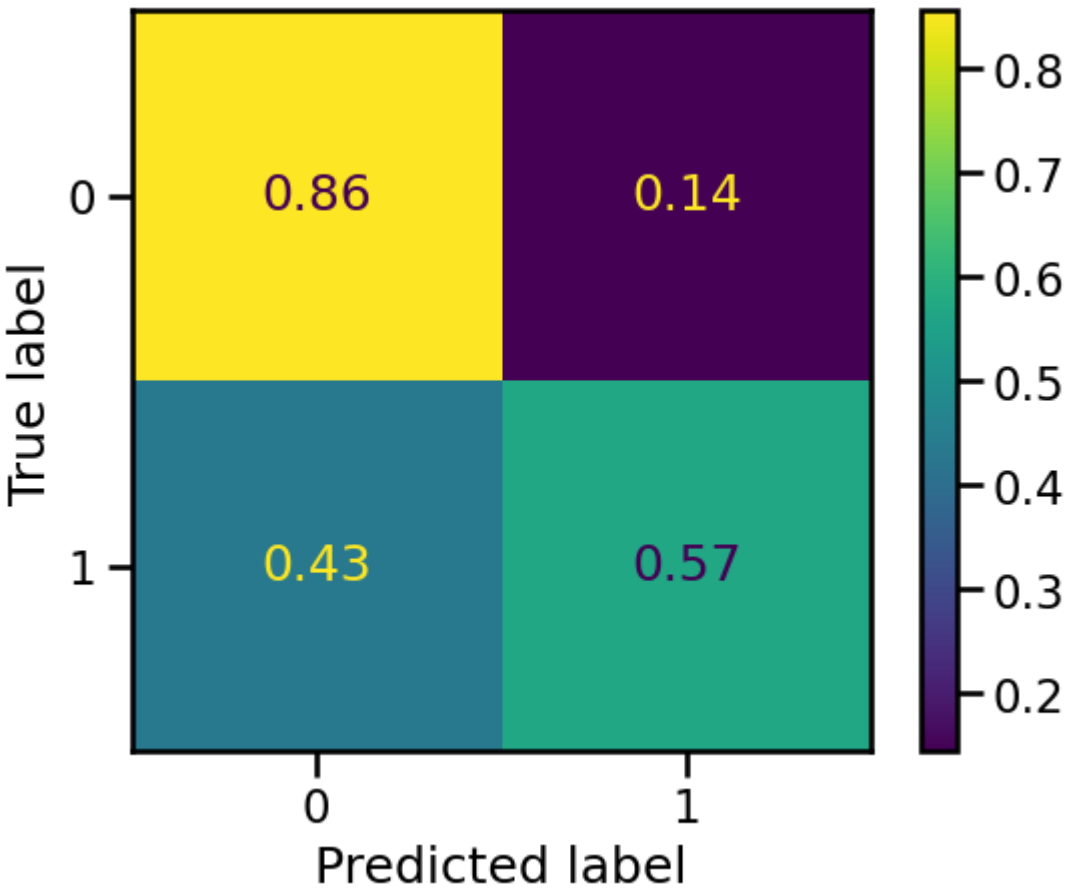
```
# Train and evaluate the tuned KNN model.
KNN_model_q.fit(X_train_scaled, y_train)

y_pred = KNN_model_q.predict(X_test_scaled)
y_pred_proba = KNN_model_q.predict_proba(X_test_scaled)[:, 1]

scores_KNN_q = get_metrics(y_test, y_pred, y_pred_proba)
print(scores_KNN_q)
```

```
Confusion Matrix:
[[10839  1836]
 [ 2080  2725]]

Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.86      0.85     12675
           1       0.60      0.57      0.58      4805

    accuracy                           0.78     17480
   macro avg       0.72      0.71      0.71     17480
weighted avg       0.77      0.78      0.77     17480

{'accuracy': 0.7759725400457665, 'precision': 0.5974566980925236, 'recall':
0.5671175858480749, 'f1_score': 0.581891949604954, 'roc_auc': 0.8049794038507
```

In [186…

```
# Plot the normalized confusion matrix for the tuned KNN model.
cf_KNN_q = confusion_matrix(y_test, y_pred, normalize='true')
```

```
sns.set_context('talk')
disp = ConfusionMatrixDisplay(confusion_matrix=cf_KNN_q, display_labels=K
disp.plot()
plt.show()
```



Decision Tree Classifier

In [187…

```
# Define the Decision Tree classifier.
from sklearn.tree import DecisionTreeClassifier
DT_model = DecisionTreeClassifier(random_state=42)
```

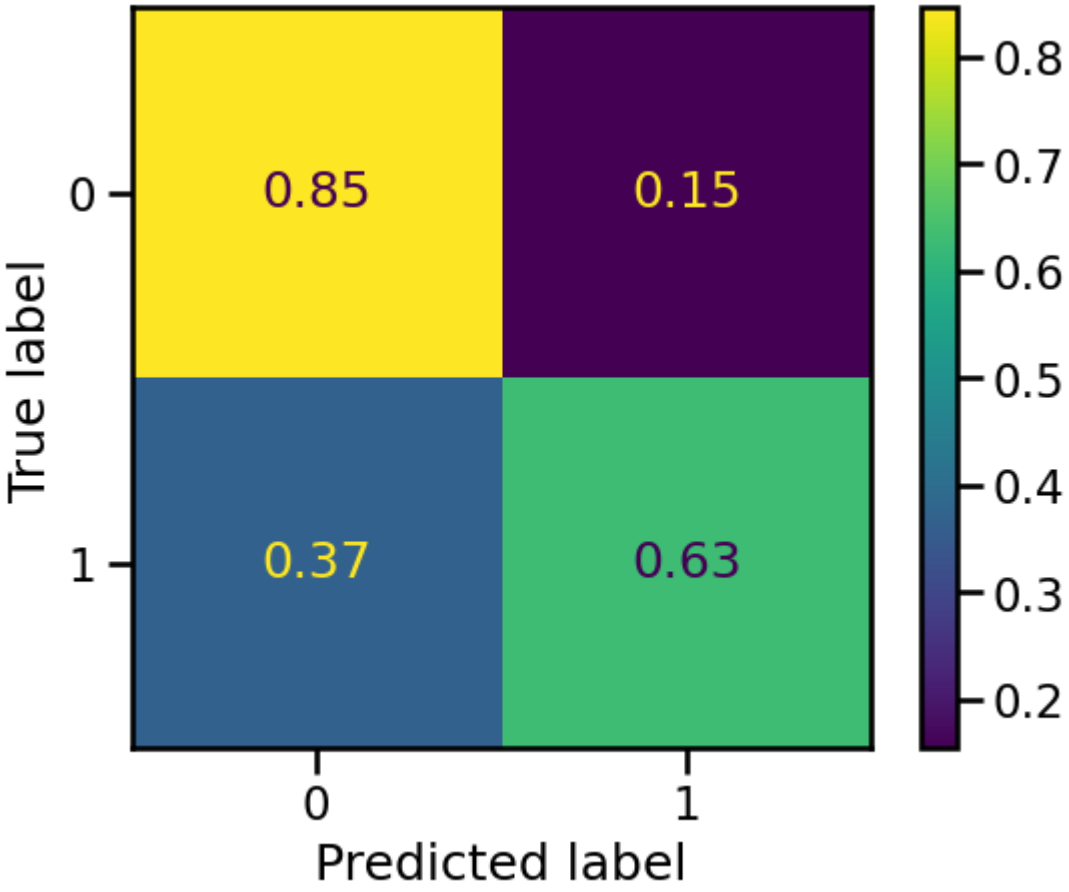
In [188…

```
# Train and evaluate the Decision Tree model.
DT_model.fit(X_train, y_train)

y_pred = DT_model.predict(X_test)
y_pred_proba = DT_model.predict_proba(X_test)[:, 1]
scores_DT = get_metrics(y_test, y_pred, y_pred_proba)
print(scores_DT)
```

```
Confusion Matrix:
[[10713  1962]
 [ 1777  3028]]

Classification Report:
              precision    recall  f1-score   support

           0       0.86      0.85      0.85     12675
           1       0.61      0.63      0.62      4805

    accuracy                           0.79     17480
   macro avg       0.73      0.74      0.73     17480
weighted avg       0.79      0.79      0.79     17480

{'accuracy': 0.7860983981693364, 'precision': 0.606813627254509, 'recall':
0.6301768990634755, 'f1_score': 0.618274629913221, 'roc_auc': 0.7391088178610
```

In [189…

```
# Plot the normalized confusion matrix for Decision Tree.
D = confusion_matrix(y_test, y_pred, normalize='true')


sns.set_context('talk')
disp = ConfusionMatrixDisplay(confusion_matrix=D, display_labels=DT_model
disp.plot()
plt.show()
```



Random Forest Classifier

In [199…

```
# Train the baseline Random Forest model.
from sklearn.ensemble import RandomForestClassifier
```

```
RF_model = RandomForestClassifier(max_depth=5, random_state=42, n_jobs=-1
RF_model.fit(X_train, y_train)
y_pred = RF_model.predict(X_test)
```

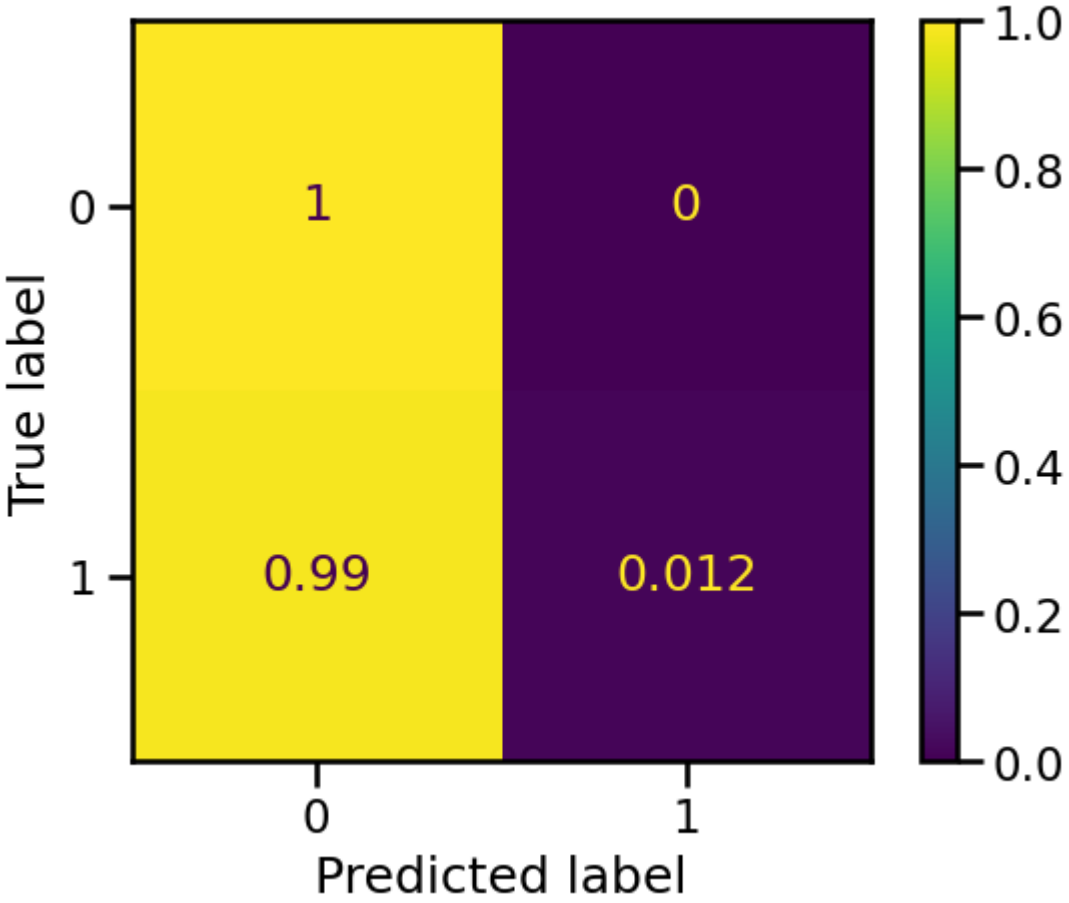
In [191…

```
# Evaluate the baseline Random Forest model.
y_pred_proba = RF_model.predict_proba(X_test)[:, 1]
scores_RF = get_metrics(y_test, y_pred, y_pred_proba)
print(scores_RF)
```

```
Confusion Matrix:
[[12675     0]
 [ 4748    57]]

Classification Report:
              precision    recall  f1-score   support

           0       0.73      1.00      0.84     12675
           1       1.00      0.01      0.02      4805

    accuracy                           0.73     17480
   macro avg       0.86      0.51      0.43     17480
weighted avg       0.80      0.73      0.62     17480

{'accuracy': 0.72837528604119, 'precision': 1.0, 'recall': 0.0118626430801248
'f1_score': 0.023447141094199916, 'roc_auc': 0.8246045395677989}
```

In [192…

```
# Plot the normalized confusion matrix for the baseline Random Forest.
F = confusion_matrix(y_test, y_pred, normalize='true')


sns.set_context('talk')
disp = ConfusionMatrixDisplay(confusion_matrix=F, display_labels=RF_model
disp.plot()
plt.show()
```



In [193…

```
# Import GridSearchCV and define a Random Forest setup for tuning.
from sklearn.model_selection import GridSearchCV

RF_model_param = RandomForestClassifier(
    random_state=42,
    n_jobs=-1,
    class_weight='balanced'
)

param_grid = {
    'n_estimators': [100, 200, 300],
    'max_depth': [5, 10, 15, None],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 3, 5],
    'max_features': ['sqrt', 'log2']
}

Grid_RF = GridSearchCV(
    estimator=RF_model_param,
    param_grid=param_grid,
    scoring='f1',
    cv=5,
    n_jobs=-1,
    verbose=1
)

Grid_RF.fit(X_train, y_train)

print("Best params:", Grid_RF.best_params_)
print("Best CV score:", Grid_RF.best_score_)
```

```
Fitting 5 folds for each of 216 candidates, totalling 1080 fits
Best params: {'max_depth': None, 'max_features': 'sqrt', 'min_samples_leaf':
'min_samples_split': 10, 'n_estimators': 300}
Best CV score: 0.7143991357268237
```

In [194…

```python
# Train and evaluate the tuned Random Forest model with chosen hyperparam
RF_model_param = RandomForestClassifier(max_depth=None, random_state=42,
RF_model_param.fit(X_train, y_train)
y_pred = RF_model_param.predict(X_test)
y_pred_proba = RF_model_param.predict_proba(X_test)[:, 1]
scores_RF_param = get_metrics(y_test, y_pred, y_pred_proba)
print(scores_RF_param)
```

```
Confusion Matrix:
[[11857   818]
 [ 1973  2832]]

Classification Report:
              precision    recall  f1-score   support

           0       0.86      0.94      0.89     12675
           1       0.78      0.59      0.67      4805

    accuracy                           0.84     17480
   macro avg       0.82      0.76      0.78     17480
weighted avg       0.83      0.84      0.83     17480

{'accuracy': 0.8403318077803204, 'precision': 0.7758904109589041, 'recall':
0.5893860561914672, 'f1_score': 0.66989946777055, 'roc_auc': 0.90200225356969
```

Gradient Boosting Classifier

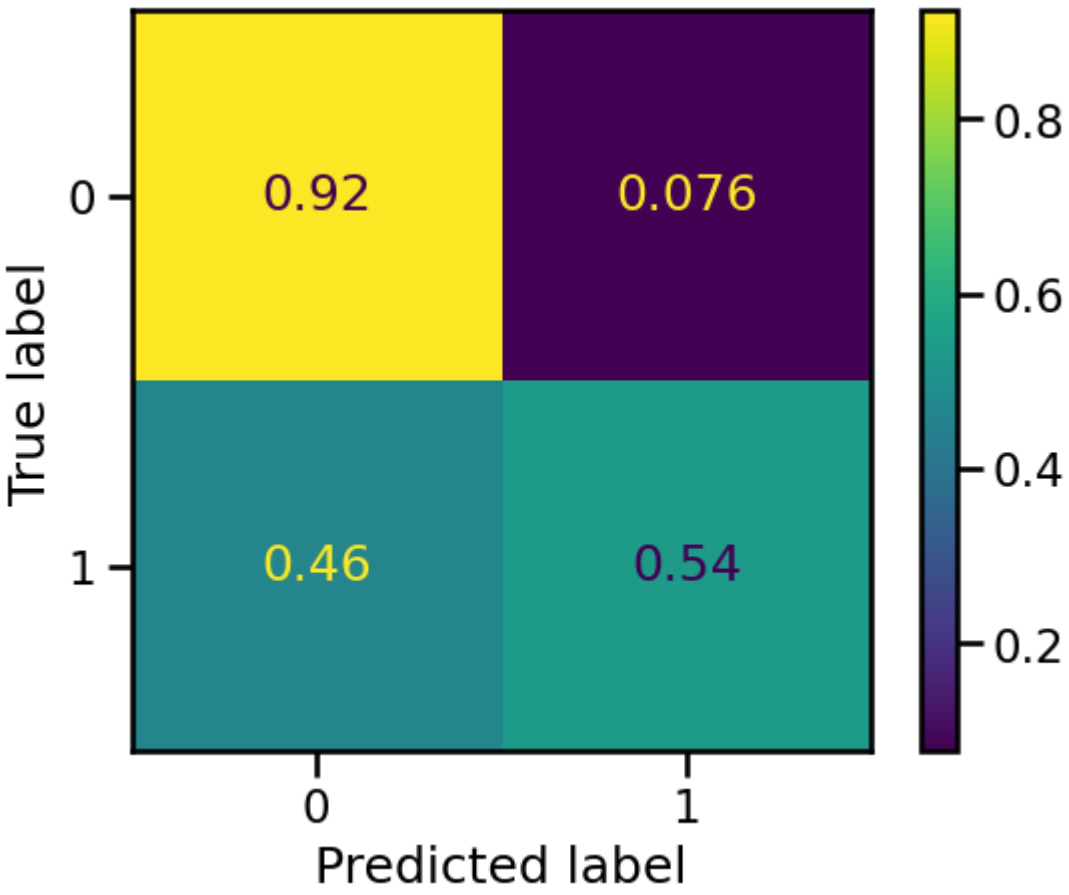
In [195…

```python
# Train and evaluate the Gradient Boosting classifier.
from sklearn.ensemble import GradientBoostingClassifier

GB_model = GradientBoostingClassifier(
    n_estimators=200,
    learning_rate=0.05,
    max_depth=3,
    min_samples_split=10,
    min_samples_leaf=5,
    subsample=0.8,
    random_state=42
)

GB_model.fit(X_train, y_train)

y_pred = GB_model.predict(X_test)
y_pred_proba = GB_model.predict_proba(X_test)[:, 1]

scores_GB = get_metrics(y_test, y_pred, y_pred_proba)
print(scores_GB)
```

```
Confusion Matrix:
[[11707   968]
 [ 2212  2593]]

Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.92      0.88     12675
           1       0.73      0.54      0.62      4805

    accuracy                           0.82     17480
   macro avg       0.78      0.73      0.75     17480
weighted avg       0.81      0.82      0.81     17480

{'accuracy': 0.8180778032036613, 'precision': 0.7281662454366751, 'recall':
0.5396462018730489, 'f1_score': 0.6198900310781735, 'roc_auc': 0.874398331783
```

In [196…

```
# Plot the normalized confusion matrix for Gradient Boosting.
G = confusion_matrix(y_test, y_pred, normalize='true')


sns.set_context('talk')
disp = ConfusionMatrixDisplay(confusion_matrix=G, display_labels=GB_model
disp.plot()
plt.show()
```



Model Comparison and Selection

To evaluate the performance of the models fairly, several metrics were compared, including accuracy, preci
recall, F 1 -score, and ROC-AUC. Because cancellation prediction is an imbalanced classification problem,
score and recall are especially important when selecting the most useful model.

In [197…

```
# Compare all trained models in one table and visualize their F1 scores.
comparison_df = pd.DataFrame([
    {'Model': 'Logistic Regression', **scores_LR},
    {'Model': 'Linear SVC', **scores_SVC},
    {'Model': 'KNN (default)', **scores_KNN},
    {'Model': 'KNN (weighted)', **scores_KNN_q},
    {'Model': 'Decision Tree', **scores_DT},
    {'Model': 'Random Forest (baseline)', **scores_RF},
    {'Model': 'Random Forest (tuned)', **scores_RF_param},
    {'Model': 'Gradient Boosting', **scores_GB}
])

comparison_df = comparison_df.sort_values(by='f1_score', ascending=False)
display(comparison_df)

plt.figure(figsize=(10, 5))
sns.barplot(data=comparison_df, x='f1_score', y='Model')
plt.title('Model Comparison by F1 Score')
plt.xlabel('F1 Score')
plt.ylabel('Model')
plt.xlim(0, 1)
plt.tight_layout()
plt.show()
```

Out[197…

	Model	accuracy	precision	recall	f 1 _sc
6	Random Forest param	0.840332	0.775890	0.589386	0.669889
7	Gradient Boosting	0.818078	0.728166	0.539646	0.619899
4	Decision Tree	0.786098	0.606814	0.630177	0.618277
3	KNN	0.775973	0.597457	0.567118	0.581889
2	KNN	0.772769	0.594509	0.545265	0.568825
0	Logistic Regression	0.794108	0.675291	0.483455	0.563499
1	Linear SVC	0.793822	0.686086	0.460770	0.551299
5	Random Forest	0.728375	1.000000	0.011863	0.023444

Final Model Choice and Conclusion

Based on the comparison above, the tuned Random Forest model was selected as the best final model for t because it achieved the highest F 1 -score and the strongest overall balance between precision and recall. important because the dataset is imbalanced, so accuracy alone would not be enough to evaluate performa

Final selected model

- Model: Random Forest (tuned)
- Why selected: Best overall classification performance on the test set, especially for identifying cancele bookings while still keeping good precision.

Key findings

- Features related to lead time, deposit type, market segment, ADR, and room/reservation characteristic an important role in predicting cancellations.
- The baseline Random Forest with shallow depth underperformed badly on recall, which shows that hyperparameter tuning was necessary.
- Simpler linear models gave reasonable results, but tree-based models captured the complex patterns i more effectively.

Final conclusion

This project demonstrates that supervised machine learning can be used successfully to predict hotel booki cancellations. After comparing multiple classification models, the tuned Random Forest provided the best l performance and practical usefulness.

These results suggest that hotels can use booking details and customer behavior variables to identify reserv with a higher cancellation risk. This kind of prediction can support better planning, pricing, and resource

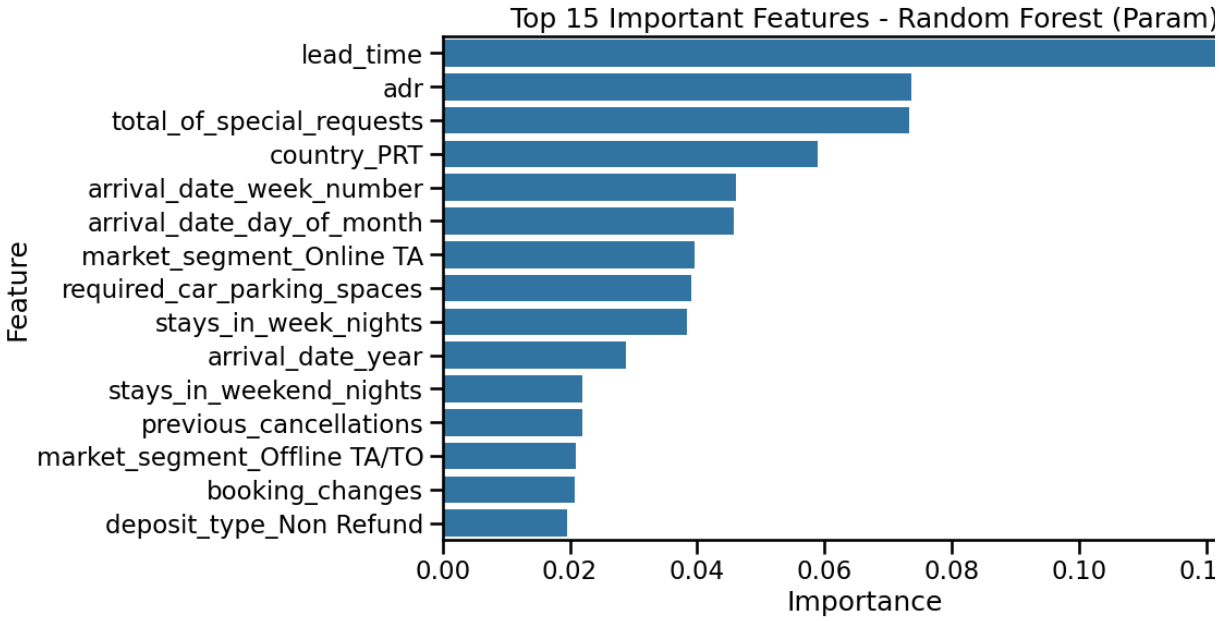
Feature Importance of the Best Model

Feature importance was used to interpret the tuned Random Forest model and identify which variables co most to prediction. This step is valuable because it connects model performance to meaningful business in

In [198…

```
# Plot the top 15 most important features from the tuned Random Forest mo
feature_importance = pd.Series(RF_model_param.feature_importances_, index
top_features = feature_importance.sort_values(ascending=False).head(15)

plt.figure(figsize=(10,6))
sns.barplot(x=top_features.values, y=top_features.index)
plt.title('Top 15 Important Features - Random Forest (Param)')
plt.xlabel('Importance')
```

```
plt.ylabel('Feature')
plt.show()
```



Submission Note

This notebook documents the full workflow required for the final project:

- dataset understanding
- preprocessing and cleaning
- exploratory data analysis
- model training and evaluation
- model comparison
- final model selection and conclusion

The notebook is structured so it can be exported as PDF for submission.